FORM HS-02 · REV. 4 POST-STASIS REFRESHER · MODULE 2 OF 9 POST AT EVERY PPE ISSUE POINT

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Health & Safety

BREATHE CLEAN.

THE SAND UNDER YOUR BOOTS IS GROUND GLASS FOR YOUR LUNGS. DUST YOU CAN'T SEE DOES THE DAMAGE.

THE SAND IS SILICA

SILICOSIS HAS NO CURE.

This world's sand is rich in crystalline silica. Cutting, grinding, drilling, blasting or just driving over it throws up **respirable dust**, particles small enough to reach deep into your lungs. Your lungs scar around them, and the scarring never heals. It keeps getting worse after exposure stops.

SILICOSIS	LUNG CANCER	COPD	KIDNEY DISEASE
Scarred, stiff lungs. Shortness of breath that never goes away. Can kill.	Crystalline silica is a known human carcinogen.	Chronic bronchitis and emphysema. Every breath is work.	Silica exposure is linked to kidney damage.

THE LIMITS

RESPIRABLE CRYSTALLINE SILICA
8-HOUR AVERAGE

50 µg/m³	25 µg/m³	Invisible
EXPOSURE LIMIT (PEL)	ACTION LEVEL: MONITORING AND MEDICAL CHECKS BEGIN	RESPIRABLE SILICA IS TOO FINE TO SEE. VISIBLE DUST MEANS FAR MORE IS THERE.

The Atmospheric Office samples dust at every active work face. Results post at muster. If a reading tops the action level, your crew goes on the medical surveillance list.

CONTROLS FIRST

RESPIRATORS ARE THE LAST LINE OF DEFENSE

1. **Wet it.** Cut, grind, core and drill with water fed to the blade or bit. Wet sweep, never dry sweep. Never clear dust with compressed air.
2. **Vacuum it.** Tools with shrouds connected to a HEPA dust collector. Clean up with a HEPA vacuum.
3. **Pull it away.** Local exhaust at fixed cutting and grinding stations in the Fabrication Yards.
4. **Enclose yourself.** Equipment cabs sealed, filtered and pressurized. Keep the doors shut.
5. **Then wear a respirator** sized for the job. See the ladder below.

RESPIRATOR LADDER

ASSIGNED PROTECTION FACTOR (APF)
HIGHER = CLEANER AIR INSIDE

N95 filtering facepiece Disposable. Short, light dust tasks.	10 × CLEANER
Half-mask elastomeric P100 cartridges. Routine cutting and grinding.	10 × CLEANER
Loose-fitting PAPR Powered hood. Works with beards. Cool airflow.	25 × CLEANER
Full-facepiece elastomeric P100. Heavy dust, protects eyes too.	50 × CLEANER
Tight-fitting full-face PAPR Abrasive blasting, sandstorm work, highest dust.	1000 × CLEANER

DO

- **Get fit-tested** on your exact model and size. Repeat every year and whenever your face changes.
- **Seal check every time** you put it on: cover the cartridges and inhale. The mask should pull in.
- **Be clean-shaven** where the seal touches your face. Beards go under a loose PAPR hood only.
- Change cartridges when breathing gets hard or on the schedule your lead posts.

DON'T

- Wear a mask you weren't fit-tested on.
- Wear a tight-fitting mask over stubble or a beard.
- Hang it around your neck between cuts. The inside fills with dust.
- Use a dust mask for gas, fumes or low oxygen. Different hazard (Modules 5, 8, S-A).

EYES

- Safety glasses marked **ANSI Z87.1** (look for "Z87" on the frame) at all times on site.
- **Sealed goggles** whenever sand is blowing, when cutting or grinding, and outside a cab in wind over 15 mph.
- Face shield **over** glasses or goggles for grinding and chemical pours. A shield alone isn't eye protection.
- Sand in your eye: don't rub. Flush with the eyewash station for at least 15 minutes, then go to the FAS.

HEARING

85	90
DBA ACTION LEVEL	DBA 8-HR LIMIT

- Old Earth rules allow 90 dBA for 8 hours, halving the time for every 5 dB louder. The stricter health guideline is 85 dBA, halving every **3 dB**. This site follows the stricter one.
- **Rule of thumb:** if you have to shout to be heard at arm's length, you need protection.
- Plugs rolled tight and fully in, or muffs sealed over your ears. Double up near chipping guns, generators and impact tools.
- Hearing loss from noise is permanent. Ringing ears after a shift means damage already happened.

PPE BY TRADE

● ALWAYS · ○ FOR THE LISTED TASKS

TRADE	HELMET	Z87 EYES	GOGGLES / SHIELD	RESPIRATOR	HEARING	GLOVES	OTHER
Welder	●	●	●	●	○	●	Welding helmet, FR leathers (Module 5)
Machinist	○	●	○	○	●	○	No gloves at rotating spindles. No loose sleeves or jewelry (Module 7)
Equipment operator	●	●	○	○	●	●	Sealed cab. Respirator if cab filter fails.
Rigger / signal	●	●	○	○	○	●	Hi-vis vest, orange helmet (Module 6)
Polymer tech	●	●	●	●	○	●	Chemical gloves and suit per the SDS (Module 8)
Laborer	●	●	○	○	○	●	N95 minimum for dry sweeping and demolition
Electrician	●	●	○	○	○	●	Class E helmet, arc-rated clothing, insulated gloves (Module 7)

Boots: safety-toe, puncture-resistant, heat-rated soles for everyone. Sand surfaces pass 150°F by midday.

SANDSTORM

Continuous long siren, or a wall of dust on the horizon. Visibility can drop to zero in minutes. Wind-driven sand strips skin and chokes filters.

MAKE SAFE. GET SEALED IN.

Ch 16 Atmospheric broadcasts storm warnings. Keep it on scan when you're working outside.

1. **Make your work safe** in seconds: torch off, load landed, machine parked with the bucket down.
2. **Goggles and respirator on** before you move. Full-face if you have it.
3. **Go to the nearest sealed shelter** with your buddy. Every grid block has one. Don't try to reach the muster point.
4. Caught in the open? **Get low behind solid cover**, back to the wind, face covered. Stay put. Call Ch 1 with your location.
5. **Head count** in the shelter. Report missing crew on Ch 1. Leave only on the all-clear.

STOP-WORK AUTHORITY

Every worker on this site can stop work when dust is visible in the air, a dust collector is down, or a crew member has no respirator for the job. Wet it, fix it, then restart.

WORKER ACKNOWLEDGEMENT

I have been fit-tested on my respirator, I know how to seal-check it, I know the controls for silica dust, and I know the sandstorm procedure and my nearest shelter.

NAME	COLONIST ID	CREW / SECTOR	DATE

Adapted from Old Earth standards: OSHA 29 CFR 1926.1153 respirable crystalline silica, OSHA 1910.134 respiratory protection (APFs), NIOSH silica and noise criteria (REL 85 dBA, 3 dB exchange), OSHA 1926.52 / 1910.95 occupational noise, ANSI/ISEA Z87.1 eye protection.